

Beamer⁺

Interactive Presentations

a guided tour

press  to begin

what's inside

Beamer⁺ turns a static PDF into a live, interactive lecture
every tile below has a demo later in this deck, so press  to explore



video & audio



3D models



split view



reveal steps



ink & shapes



magnifier



live widgets

```
>>>_
```

Python



live polls



scratch paper



screen recording

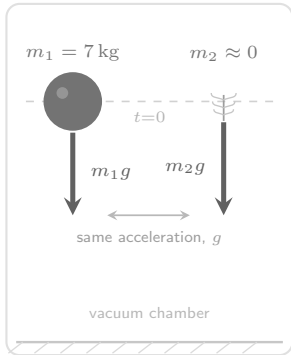


notes PDF

videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2nd law in a Physics course



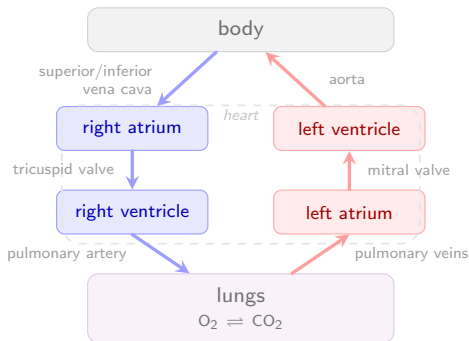
gravitational acceleration is
independent of mass



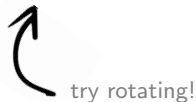
models

supported formats include .glpb

example usage: explaining blood flow through the heart in an Anatomy course



blood flow through the heart
(deoxygenated blood / oxygenated blood)



reveal steps

uncover a slide one piece at a time, with no \pause or recompiling
example usage: walking through a proof in a Discrete Math class

Claim. $\sqrt{2}$ is irrational.

- 1 Suppose not: $\sqrt{2} = \frac{p}{q}$ with p, q integers in *lowest terms*.
- 2 Squaring, $p^2 = 2q^2$, so p^2 is even $\Rightarrow p$ is even: $p = 2k$.
- 3 Then $4k^2 = 2q^2$, so $q^2 = 2k^2 \Rightarrow q$ is even too.
- 4 So p and q are both even, and share a factor of 2.

Contradiction! Lowest terms can't both be even. ■

 press  to reveal!

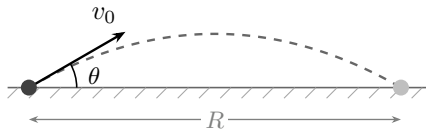
ink & annotations

pens, highlighters, shapes, $\text{\LaTeX}$ text boxes, a spotlight and a magnifier
example usage: working a projectile problem live in a Physics class

A ball is kicked at $v_0 = 20 \text{ m/s}$, at $\theta = 30^\circ$ above level ground. How far away does it land?

The range is

$$R = \frac{v_0^2 \sin 2\theta}{g} = \frac{(20)^2 \sin 60^\circ}{9.81}$$



Psst, you found the fine print! Scroll while the magnifier is on to zoom anywhere from 1.5x to 5x. It works on ink, videos and widgets too.



try the magnifier on the fine print!

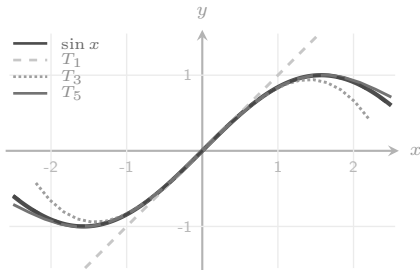
split view

allows two slides to be shown simultaneously

example usage: explaining Taylor series in a Calculus class

Any smooth function f can be approximated around $x_0 = 0$ as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



expansion of $f(x) = \sin(x)$ around $x_0 = 0$

function plotter

2D curves and 3D surfaces you can rotate, zoom and edit live
example usage: classifying critical points in a Multivariable Calculus class

At a critical point, $\nabla f = \mathbf{0}$. The
second-derivative test looks at

$$D = f_{xx}f_{yy} - f_{xy}^2.$$

For $f(x, y) = ax^2 - y^2$ at the origin:

$$D = (2a)(-2) - 0^2 = -4a$$

$a > 0$: a **saddle point**

$a < 0$: a local maximum

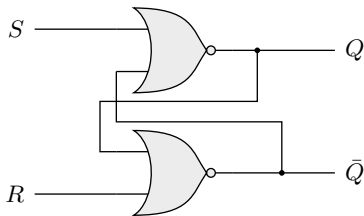
 try dragging the slider for a !

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates
inputs are *active-high*

S	R	Q	$\bar{Q}$	state
0	0	Q	$\bar{Q}$	hold
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	forbidden

truth table for an SR latch
implemented with NOR gates

record & share

the lecture doesn't have to stay in the room

example usage: posting the recording and annotated notes right after class



record the tab

slides, ink, widgets
and sound, saved
every second so a
crash loses nothing



notes PDF

one handout with
your ink, every wid-
get as it stands,
and the poll results



one ZIP file

slides, media, widgets,
ink and poll setup
together, ready to
reopen next term

try the record button!



make your own

start from the PDF you already have
everything you've seen in this deck takes three steps

1



load your slides

drop a PDF onto the
window, or a Beamer⁺
ZIP you saved earlier

2



add the extras

open edit mode to place
videos, 3D models,
widgets and polls

3



present

draw, poll and record
live, then download
the ZIP and your notes